

PART-B

Name of the Student: SATYAKI RAY

Signature of the Student: Satyaki Ray

Roll No.: 230123080

Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-B of the examination has **4 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No questions require any clarification from the instructor or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	4	Total
Points:	6	7	5	2	20
Score:	1	7	4	2	14

QUESTIONS

1. (6 points) Let

$$S = \begin{pmatrix} S_{yy} & S_{yX}^T \\ S_{yX} & S_{XX} \end{pmatrix}$$

be the partitioned sample covariance matrix of $Y_{1 \times 1}$ and $X_{q \times 1}$ based on a centered (with respect to sample mean) sample of size n . Prove that the squared sample first canonical correlation r_1^2 is identical to the coefficient of determination (R^2) obtained from the multiple linear regression of Y on the vector X .

$\Rightarrow r_1^2 =$ the largest eigenvalue of

$$S_{yy}^{-\frac{1}{2}} S_{yx} S_{xx}^{-1} S_{yx}^T S_{yy}^{-\frac{1}{2}}$$

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subject all dot product. In particular, let $\mathbf{u} = (1, 0, 0)$ and $\mathbf{v} = (0, 1, 0)$. Then $\mathbf{u} \cdot \mathbf{v} = 0$, so $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

For any vector $\mathbf{x} = (x_1, x_2, x_3)$, we can write $\mathbf{x}$ as a linear combination of $\mathbf{u}, \mathbf{v}, \mathbf{w}$. For example, $\mathbf{x} = x_1\mathbf{u} + x_2\mathbf{v} + x_3\mathbf{w}$. This shows that $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is a basis for $\mathbb{R}^3$. Since $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are orthogonal, this is an orthonormal basis.

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

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2. Suppose you are performing a factor analysis on a 4×4 sample covariance matrix $\mathbf{S}$. The eigenvalues of $\mathbf{S}$ are $\lambda_1 = 0.8$, $\lambda_2 = 2.1$, $\lambda_3 = 4.2$, and $\lambda_4 = 0.2$. The corresponding normalized eigenvectors are $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and $\mathbf{e}_4$. You decide to extract $m = 2$ factors using the principal component method.
- (a) (2 points) Write down the explicit mathematical expression for the estimated factor loading matrix $\tilde{\mathbf{L}}$ in terms of the given eigenvalues and eigenvectors.
- (b) (1 point) Let the first row of $\tilde{\mathbf{L}}$ be $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$. Calculate the estimated communality $\tilde{h}_1^2$ for the first variable.
- (c) (4 points) Prove that the sum of all estimated communalities $\sum_{i=1}^p \tilde{h}_i^2$ is exactly equal to the sum of the largest $m = 2$ eigenvalues of $\mathbf{S}$.

$\Rightarrow$ (a) Using the principal component method, the loading matrix $\tilde{\mathbf{L}}$ is given as

$$\tilde{\mathbf{L}} = [\sqrt{\hat{\lambda}_1} \hat{\mathbf{e}}_1 \mid \sqrt{\hat{\lambda}_2} \hat{\mathbf{e}}_2 \mid \dots \mid \sqrt{\hat{\lambda}_m} \hat{\mathbf{e}}_m]$$

where, $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots \geq \hat{\lambda}_m$ are eigenvalues of $\mathbf{S}$.

Here, $m = 2$, so the two largest eigenvalues are

$\lambda_3 = 4.2$ and $\lambda_2 = 2.1$, with their corresponding eigenvectors being $\mathbf{e}_3$ and $\mathbf{e}_2$.

$$\therefore \tilde{\mathbf{L}} = [\sqrt{4.2} \mathbf{e}_3 \mid \sqrt{2.1} \mathbf{e}_2]$$

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$$= [2.0494 \mathbf{e}_3 \mid 1.4491 \mathbf{e}_2]$$

(b) First row, $\tilde{\mathbf{l}}_1^T = [0.8, -0.4]$

$$\begin{aligned} \therefore \text{Communality, } \tilde{h}_1^2 &= \tilde{l}_{11}^2 + \tilde{l}_{12}^2 \\ &= (0.8)^2 + (-0.4)^2 \\ &= 0.8 \end{aligned}$$

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(c) Let $\mathbf{e}_3 = \begin{bmatrix} e_{31} \\ e_{32} \\ e_{33} \\ e_{34} \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} e_{21} \\ e_{22} \\ e_{23} \\ e_{24} \end{bmatrix}$

The eigenvectors are normalized.

$$\left. \begin{aligned} e_{31}^2 + e_{32}^2 + e_{33}^2 + e_{34}^2 &= 1 \\ e_{21}^2 + e_{22}^2 + e_{23}^2 + e_{24}^2 &= 1 \end{aligned} \right\} \quad (*)$$

Here, $p=4$

Now, $\sum_{i=1}^4 \tilde{h}_i^2 = \sum_{i=1}^4 (\tilde{l}_{i1}^2 + \tilde{l}_{i2}^2)$

Now, $\tilde{L} = \begin{bmatrix} 2.0494 e_{31} & \sqrt{4.2} e_{31} & \sqrt{2.1} e_{21} \\ 2.0494 e_{32} & \sqrt{4.2} e_{32} & \sqrt{2.1} e_{22} \\ 2. & \sqrt{4.2} e_{33} & \sqrt{2.1} e_{23} \\ & \sqrt{4.2} e_{34} & \sqrt{2.1} e_{24} \end{bmatrix}$

$$\begin{aligned} \sum_{i=1}^4 (\tilde{l}_{i1}^2 + \tilde{l}_{i2}^2) &= 4.2 e_{31}^2 + 2.1 e_{21}^2 \\ &\quad + 4.2 e_{32}^2 + 2.1 e_{22}^2 \\ &\quad + 4.2 e_{33}^2 + 2.1 e_{23}^2 \\ &\quad + 4.2 e_{34}^2 + 2.1 e_{24}^2 \\ &= 4.2 (e_{31}^2 + e_{32}^2 + e_{33}^2 + e_{34}^2) \\ &\quad + 2.1 (e_{21}^2 + e_{22}^2 + e_{23}^2 + e_{24}^2) \\ &= 4.2 \times 1 + 2.1 \times 1 \quad (\text{using } (*)) \\ &= 4.2 + 2.1 \\ &= \lambda_3 + \lambda_2 \end{aligned}$$

i.e. the sum of the $m=2$ largest eigenvalues of S . (Proved).

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3. (5 points) Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a distribution with mean $\mu \in \mathbb{R}$ and variance μ^2 . Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Derive a variance-stabilizing transformation $g(\bar{X}_n)$ such that the asymptotic variance of $g(\bar{X}_n)$ does not depend on μ .

$\Rightarrow$ We know that, if $\hat{\theta}$ is an estimator of θ , and

$$\sqrt{n}(\hat{\theta} - \theta) \sim N(0, \sigma^2)$$

then by delta method, $\sqrt{n}(g(\hat{\theta}) - g(\theta)) \sim N(0, \sigma^2 [g'(\theta)]^2)$

Here, $\hat{\theta} = \bar{X}_n$ and $\theta = \mu$.

$$\text{So, } \sqrt{n}(\bar{X}_n - \mu) \sim N(0, \mu^2)$$

$$\text{So, } \sqrt{n}(g(\bar{X}_n) - g(\mu)) \sim N(0, \mu^2 [g'(\mu)]^2)$$

We need to find $g(\cdot)$ s.t. ~~$g(\bar{X}_n)$~~

~~Var $g(\bar{X}_n)$~~ asymptotic variance of $g(\bar{X}_n)$ is independent of μ .

From Taylor series expansion,

$$g(y) = g(Ey) + (y - Ey)g'(Ey) + \dots$$

$$\text{or, } \text{Var}(g(y)) = \text{Var}(y) \cdot (g'(Ey))^2$$

For $\text{Var}(g(y))$ to be independent of parameters, we need, $(g'(Ey))^2 \propto (\text{Var}(y))^{-1}$

$$\text{or, } g'(y) \propto \frac{1}{\sqrt{\text{Var}(y)}}$$

$$\text{or, } g(y) \approx \int \frac{dy}{\sqrt{\text{Var}(y)}}$$

~~If we let $g(x) = \frac{1}{x}$, then $g'(x) = -\frac{1}{x^2}$~~

~~So, asymptotic variance of $g(\bar{X}_n)$ becomes~~

$$= \mu^2 [g'(\mu)]^2 = \mu^2$$

$$\text{Now, } \text{Var}(\bar{X}_n) = \frac{\mu^2}{n} \approx \mu^2$$

$$\therefore g(\mu) = \int \frac{d\mu}{\sqrt{\mu^2}} = \int \frac{d\mu}{\mu} = \ln \mu$$

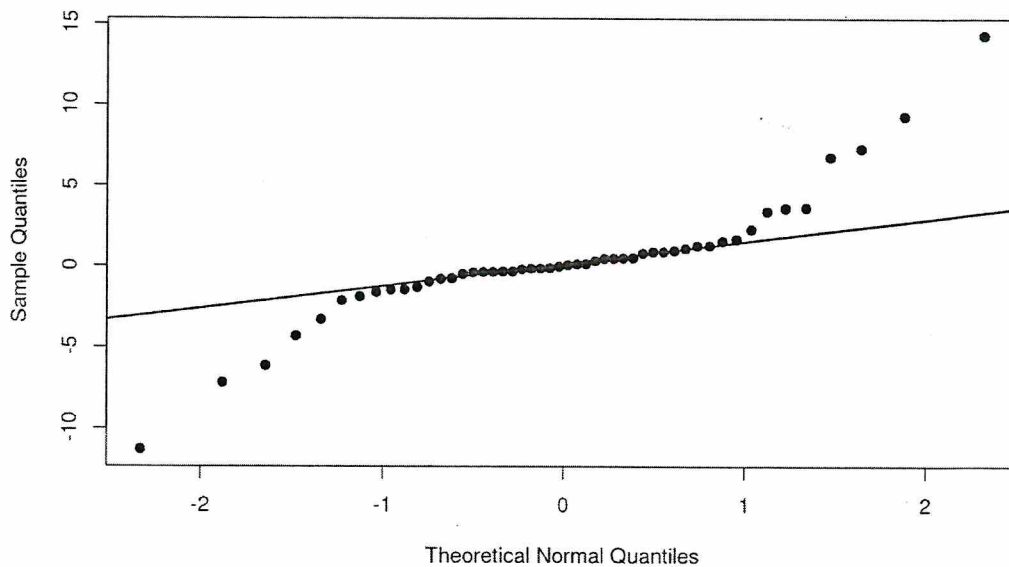
Hence an appropriate variance-stabilizing transformation is

$$\boxed{g(\bar{X}_n) = \ln \bar{X}_n} \quad \text{i.e. } g(x) = \ln x$$

$$\Rightarrow g'(x) = \frac{1}{x}$$

Then, variance = $\mu^2 [g'(\mu)]^2 = \mu^2 \cdot \frac{1}{\mu^2} = 1$ (constant independent of μ).

4. (2 points) The figure below displays a Normal Q-Q plot for a dataset of size $n = 50$. Interpret this plot to determine if the data follows a normal distribution. If the normality assumption is violated, specify which class of distributions would more accurately characterize the data.



⇒ No, the data does not follow a normal distribution. If it was normal, then the plot would be a straight line along $y = x$. But here the tails are long.

This kind of distribution is specifically called Cauchy (long-tailed) distribution. In this case, our normality assumption is invalid and we have to use some other methods like bootstrapping for estimation.

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ADDITIONAL PAGES FOR ANSWERS

For the first part of the question, we need to find the value of x such that the function $f(x) = x^2 + 2x + 1$ is a perfect square. We can rewrite the function as $f(x) = (x+1)^2$. This shows that $f(x)$ is a perfect square for all values of x . Therefore, the answer to the first part of the question is that $f(x)$ is a perfect square for all x .

For the second part of the question, we need to find the value of x such that the function $f(x) = x^2 + 2x + 1$ is a perfect square. We can rewrite the function as $f(x) = (x+1)^2$. This shows that $f(x)$ is a perfect square for all values of x . Therefore, the answer to the second part of the question is that $f(x)$ is a perfect square for all x .

